

Ride-Hailing Operational Reliability & Fare Integrity Analysis

1. Executive Summary

Analysis revealed that operational issues are strongly linked to application instability, particularly within app versions 2.3.1 and 2.4.0.

Key Observations

- Version 2.3.1 recorded the highest overall risk score (36.02).
- Version 2.4.0 recorded the second-highest risk score (33.80).
- GPS-affected trips generated a 47.32% fare error rate versus only 4.88% for trips without GPS issues.
- Versions 2.3.1 and 2.4.0 produced more than 55% fare error rates.
- Driver churn exceeded 32% on versions 2.3.1 and 2.4.0.
- Revenue at risk exceeded \$200K on version 2.3.1 alone.
- Delhi recorded the highest fare dispute concentration (21.52%).

These findings indicate that fare integrity issues and driver churn are concentrated within specific app versions rather than being uniformly distributed across the platform.

2. Business Problem

A hyper-local ride-hailing platform experienced increasing operational challenges as technical issues within the Driver App began affecting fare accuracy and driver experience. Frequent app crashes and GPS lag events created inconsistencies in trip tracking and fare calculations, making it difficult for Engineering and Operations teams to identify the root causes of fare disputes and driver churn.

Over the Last Four Months

- Driver churn increased to 24.10%.
- Fare dispute rate reached 19.98%.
- GPS lag affected 20.60% of app sessions.
- App crash rate reached 14.96%.
- Revenue at risk from disputed fares exceeded \$254K.

Business Risks

- Loss of active drivers
- Increased driver acquisition costs
- Reduced platform reliability
- Increased customer support workload
- Revenue loss due to fare disputes

The organization needed a centralized operational intelligence solution to identify unstable app versions, quantify fare integrity risks, and support technical remediation decisions.

3. Business Objectives

The objective of this analysis was to:

- Consolidate app stability and fare integrity metrics into a single dashboard.
- Identify app versions contributing most to operational risk.
- Detect device and OS combinations causing instability.
- Measure the impact of GPS issues on fare accuracy.
- Identify regions with elevated dispute rates.
- Quantify driver churn linked to technical instability.
- Support data-driven hotfix and rollback decisions.
- Reduce fare disputes and driver attrition.

4. Decision Makers

Engineering Manager

Responsible for:

- App stability monitoring
- Release quality management
- Hotfix prioritization
- Technical risk mitigation

Key Questions

- Which app versions are causing the highest instability?
- Which OS combinations generate the most crashes?
- Which devices require optimization?
- Which versions should be fixed or rolled back?

Operations Manager

Responsible for:

- Driver retention
- Fare dispute management
- Regional operational performance
- Revenue protection

Key Questions

- Why are drivers leaving the platform?
- Which cities generate the highest dispute rates?
- What technical issues are affecting fare integrity?
- Where is revenue risk concentrated?

5. Dataset Overview

The analysis was performed using:

- Driver Data
- App Session Data
- App Event Logs
- Trip Records
- Fare Dispute Data

The dataset includes:

- Driver activity
- App crashes
- GPS lag events
- Device and OS information
- Trip-level fare calculations
- Dispute resolution records

The analysis combines operational, technical, and customer-impact metrics to identify the relationship between app stability and business performance.

6. KPI Framework

Crash Rate

Percentage of app sessions experiencing crashes.

GPS Lag Rate

Percentage of sessions impacted by GPS latency issues.

Driver Churn Rate

Percentage of drivers who left the platform.

Fare Dispute Rate

Percentage of trips resulting in fare disputes.

Revenue at Risk

Estimated revenue exposure from disputed fares.

High-Risk Versions

Number of app versions exceeding stability thresholds.

Affected Sessions

Total sessions impacted by app crashes.

Fare Error Rate

Percentage of trips with incorrect fare calculations.

7. Analysis Approach

The investigation followed a structured root-cause framework.

Step 1: Validate Platform Instability

- Crash Rate Analysis
- GPS Lag Analysis
- Monthly Stability Trend Analysis

Step 2: Identify Technical Root Causes

- App Version Analysis
- OS Compatibility Analysis
- Device Stability Analysis
- Feature-Level Failure Analysis

Step 3: Measure Fare Integrity Impact

- GPS Impact on Fare Errors
- Revenue at Risk Analysis
- Fare Dispute Analysis

Step 4: Measure Driver Impact

- Driver Churn Analysis
- Regional Dispute Analysis

Step 5: Identify High-Risk Versions

- Composite Risk Score Model
- Version Prioritization Framework

8. Dashboard Overview

App Stability & Reliability Dashboard

Provides:

- Overall Crash Rate
- Overall GPS Lag Rate
- High-Risk Versions
- Affected Sessions
- Crash Rate Across App Versions
- GPS Lag Across App Versions
- Device Stability Analysis
- Crash Rate by OS Version
- High-Risk Version Detection
- Monthly Crash Trend

Purpose

Identify technical root causes driving platform instability.

Fare Integrity & Driver Impact Dashboard

Provides:

- Driver Churn Rate
- Fare Dispute Rate
- Revenue at Risk
- Total Fare Disputes
- GPS Impact on Fare Errors
- City-wise Dispute Concentration
- Driver Churn by App Version
- Fare Error vs Revenue at Risk
- GPS Lag vs Fare Error Correlation

Purpose

Measure the operational impact of technical issues on revenue and driver retention.

9. Detailed Findings

Finding 1: Versions 2.3.1 and 2.4.0 Are Driving Platform Risk

Evidence

Overall Risk Score:

- Version 2.3.1 = 36.02
- Version 2.4.0 = 33.80

All remaining versions recorded risk scores below 11.

Insight

Operational and technical issues are highly concentrated within two app releases rather than being evenly distributed across the platform.

Business Impact

A small number of unstable releases are responsible for the majority of platform reliability issues, fare disputes, and driver dissatisfaction.

Recommendation

Prioritize immediate hotfixes or controlled rollbacks for versions 2.3.1 and 2.4.0.

Finding 2: GPS Issues Are The Primary Driver Of Fare Errors

Evidence

Fare Error Rate:

- GPS Issue Trips = 47.32%
- Non-GPS Issue Trips = 4.88%

Revenue At Risk:

- GPS Issue Trips = \$447,203
- Non-GPS Issue Trips = \$16,204

Insight

GPS instability has a direct and measurable impact on fare calculation accuracy.

Business Impact

Incorrect fare calculations increase customer complaints, support costs, refund expenses, and revenue leakage.

Recommendation

Prioritize GPS stability improvements before introducing new product features.

Finding 3: Driver Churn Is Concentrated In High-Risk Versions

Evidence

Driver Churn Rate:

- Version 2.4.0 = 32.93%
- Version 2.3.1 = 32.45%

All other versions remained below 15%.

Insight

Drivers are significantly more likely to leave the platform when using unstable app versions.

Business Impact

Higher churn increases driver acquisition costs and reduces marketplace supply.

Recommendation

Focus engineering efforts on stabilizing versions associated with elevated churn rates.

Finding 4: Fare Integrity Issues Are Concentrated In Two Versions

Evidence

Fare Error Rate:

- Version 2.3.1 = 56.24%
- Version 2.4.0 = 55.57%

All remaining versions stayed below 18%.

Insight

Fare calculation problems are primarily linked to specific app releases rather than platform-wide behavior.

Business Impact

Incorrect fare calculations reduce customer trust and create financial exposure.

Recommendation

Implement mandatory fare-validation testing before future production releases.

Finding 5: Certain Device Models Experience Elevated Instability

Evidence

Highest Crash Rates:

- Realme 8 = 18.38%
- Redmi Note 10 = 16.47%
- Samsung Galaxy M31 = 16.34%

Insight

Device-specific compatibility issues contribute significantly to application instability.

Business Impact

Driver experience varies across devices, creating inconsistent platform performance.

Recommendation

Expand device compatibility testing for high-volume Android devices.

Finding 6: GPS-Related Features Generate Most Technical Events

Evidence

Highest Event Volumes:

- Navigation GPS Dropout = 1,691
- Fare Meter GPS Dropout = 1,672
- Map GPS Dropout = 1,606

Insight

GPS-related functionality represents the largest source of technical instability.

Business Impact

Navigation failures and fare tracking inaccuracies directly affect ride quality and fare integrity.

Recommendation

Prioritize engineering investment in GPS handling, location tracking, and trip monitoring components.

Finding 7: Revenue Risk Is Concentrated In High-Risk App Versions

Evidence

Revenue At Risk exceeded:

- \$200K on Version 2.3.1 alone

The majority of fare disputes originated from high-risk versions.

Insight

Financial exposure is not evenly distributed and is concentrated within a small number of unstable releases.

Business Impact

Unstable app versions directly contribute to revenue leakage and operational losses.

Recommendation

Introduce version-level revenue risk monitoring before approving production deployments.

Finding 8: App Stability Directly Influences Fare Integrity And Driver Retention

Evidence

High-risk versions simultaneously showed:

- Higher Crash Rates
- Higher GPS Lag Rates
- Higher Fare Error Rates
- Higher Driver Churn Rates

Insight

Technical instability creates a chain reaction that affects operational performance, customer experience, and driver retention.

Business Impact

Platform reliability issues have both operational and financial consequences.

Recommendation

Adopt a version-level Operational Risk Score as a release governance metric to support hotfix and rollback decisions.

10. Action Plan

Priority	Action	Owner
High	Implement immediate hotfixes for App Versions 2.3.1 and 2.4.0	Engineering Manager
High	Improve GPS stability and trip-tracking reliability	Engineering Manager
High	Introduce fare calculation validation testing before production release	Engineering Manager
High	Reduce driver churn risk through version stabilization initiatives	Operations Manager
Medium	Expand device compatibility testing for high-volume Android devices	Engineering Manager
Medium	Monitor fare dispute concentration in high-risk cities	Operations Manager

Priority	Action	Owner
Medium	Establish version-level operational risk monitoring framework	Engineering Manager
Medium	Implement release approval checks using Operational Risk Scores	Engineering Manager
Low	Develop proactive alerts for abnormal crash and GPS lag spikes	Engineering Manager
Low	Conduct periodic driver experience reviews for high-risk app versions	Operations Manager

11. Expected Business Outcomes

If recommendations are implemented:

- Reduce fare disputes by at least 25%.
- Reduce driver churn.
- Improve app stability across Android and iOS devices.
- Reduce revenue exposure from fare inaccuracies.
- Improve fare calculation reliability.
- Improve driver satisfaction and retention.
- Strengthen release governance through risk-based version monitoring.
- Improve overall platform reliability and customer trust.